HEALTHCARE · MULTI-AGENT ARCHITECTURE

Clinical Trial Matching Copilot

How three agents — and one that can say “not yet” — make an LLM workflow trustworthy.

LangGraph + **Groq · Llama-3.3-70B** + **strict JSON contracts**

Richard Abishai · Designing Agent Architectures · Live demo

The problem with one-shot trial matching

Semi-structured notes

Patient histories are messy prose, not clean fields. Key facts hide between the lines.

Brittle eligibility criteria

Long inclusion / exclusion lists where a single unmet line flips the outcome.

Missing data → false confidence

A single agent fills gaps with plausible guesses and sounds certain doing it.

The trap: one model that both reads the notes and decides eligibility is opaque — you can't see where judgment slipped in.

Three agents, three jobs

Separate gathering evidence from making the call — then let one agent govern the whole thing.

Researcher

Extracts structured facts from the patient case and trial criteria. Flags gaps.

Does NOT decide eligibility.

Evaluator

Reads only the evidence. Returns a verdict with confidence — and can refuse to guess.

The gatekeeper.

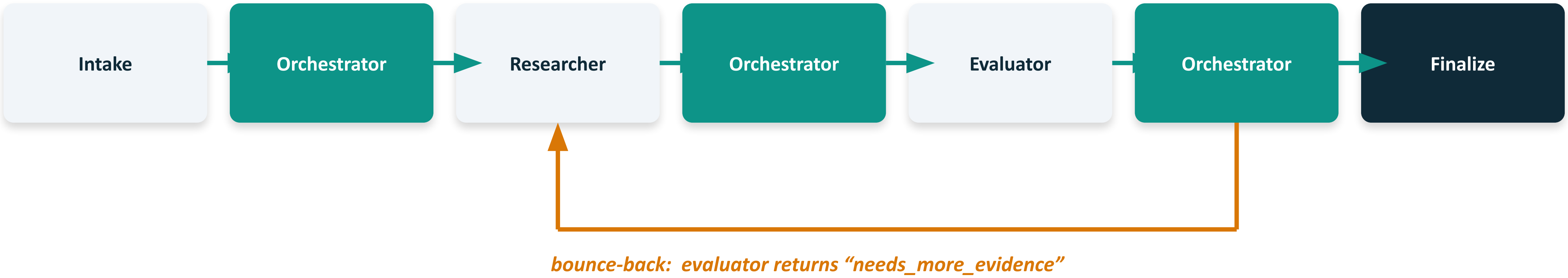
Orchestrator

Routes the flow, handles the bounce-back, terminates the loop, finalizes.

The brain.

The graph

Shared state flows through nodes. The orchestrator sits between every step.



State + nodes + edges. Every node reads & writes one shared state object, compiled before it runs.

The loop is explicit. Retry logic lives in the orchestrator, not buried in a prompt. Debuggable.

Shared state & strict output contracts

TrialState (TypedDict)

```
patient_case
trial_criteria
evidence_pack
evidence_gaps
evaluator_report
final_recommendation
next_step
retry_count
bounce_reason
```

Why strict JSON matters

Researcher →

patient_facts · trial_requirements · evidence_gaps · matching_notes

Evaluator →

verdict · confidence · exclusion_risks · questions_for_coordinator

safe_json() →

strips ```fences```, grabs the outer object, never throws mid-run

The stack — and why

LangGraph

Stateful graph orchestration. Nodes, edges, and a compiled control flow built for loops and retries.

Groq · Llama-3.3-70B

Fast inference through one clean API (langchain-groq + GROQ_API_KEY). Keeps a live demo simple.

Strict JSON contracts

TypedDict state + JSON-only agent outputs keep roles from blurring into each other.

temperature = 0 for the live run → deterministic behavior on stage.

The demo case

One realistic patient. One oncology trial. Most criteria clearly met — one is not.

Patient

- 58-year-old female
- Stage II ER-positive breast cancer
- ECOG performance status 1
- Prior chemotherapy completed 8 months ago
- **Mild liver enzyme elevation — no LFT values on file**
- No documented brain metastases

Trial criteria

INCLUSION

Age 18–70 · confirmed breast cancer · ECOG 0–1 · prior therapy allowed

EXCLUSION

Severe hepatic impairment · active CNS metastases · uncontrolled infection

The hinge: “severe hepatic impairment” can't be cleared from “mild elevation, no labs.” That ambiguity is the whole demo.

The moment: one agent catches another

Live trace, no human in the loop:

Researcher · pass 1	Extracts the facts — and is over-confident. Glosses the liver gap.
Evaluator	Returns NEEDS_MORE_EVIDENCE: “What are the current LFT values?”
Orchestrator	Hears the refusal and routes BACK to the researcher with that exact question.
Researcher · pass 2	Re-examines honestly: severity cannot be determined from the source.
Evaluator + finalize	Holds firm; orchestrator stops the loop and finalizes CONSERVATIVELY.

Delete the gatekeeper — and watch it fail

`Same patient. One line changed:  EVALUATOR_ENABLED = False`

Gatekeeper ON

POTENTIALLY ELIGIBLE — not safe to finalize.

Surfaces the open question. Asks for the LFT values. Refuses to overclaim.

Gatekeeper OFF

LIKELY ELIGIBLE. Next: coordinator review before screening.

No caveat. No question. Confidently wrong on a patient whose exclusion was never cleared.

Remove one agent and the system becomes confident — and unsafe. That gap is the value of the design.

What travels to your domain

Separate evidence from judgment

Make retries explicit & debuggable

Let agents refuse to guess

Same graph, new domain

Fintech

Loan / KYC diligence: extract ⟶ assess risk ⟶ route exceptions.

Edtech

Student intervention triage: gather signals ⟶ evaluate ⟶ escalate.

Scientific R&D

Paper screening: extract claims ⟶ verify ⟶ synthesize.

The repo is yours — fork it. Building agents? Let's talk. — Richard Abishai

Not a clinical decision system — an architecture demo. Real deployment needs validation, audit & oversight.